# Sameer Baskota

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Lalitpur, Nepal

## EDUCATION

**IOE, Pulchowk Campus** 2022 - 2026
*Bachelor in Aerospace Engineering — Location: Lalitpur, Nepal*

## EXPERIENCE

**Research Assistant**
*Geovation Nepal* *July 2025- Present*

- Conduct field operations for drone-based precision seeding in agricultural fields.
- Investigate factors affecting seeding efficiency, including payload capacity, flight parameters, and seed distribution mechanisms
- Study the challenges and opportunities of adopting agricultural drones in Nepal.
- Investigate ways to improve the performance, reliability, and efficiency of agricultural drones for use in Nepal.

**Internship - AMO (Yeti Airlines)** Nov 2025 - Jan 2026

- Supported aircraft maintenance operations through coordination with MCC and AMO teams
- Assisted in maintenance documentation, tooling, and spare-parts inventory management
- Participated in Quality Assurance and production-planning related audit activities

**Flying Club, Pulchowk Campus** Dec 2024 - Jan 2025

- Participated in UAV design, fabrication, and flight testing
- Gained hands-on experience with RC aircraft assembly, drone systems, and flight operations
- Worked with a multidisciplinary team on aerodynamic design and flight control systems

## PROJECTS

**UAV-Based Crop Health Monitoring Using RGB Based Imaging** Aug 2025 - Mar 2026
*Tools: Python, OpenCV, Scikit-learn, DJI Phantom 4, QGIS, DroneDeploy*

- Developed a UAV-based system for maize crop health monitoring
- Generated orthomosaic maps from RGB aerial imagery
- Extracted vegetation indices (ExG, VARI, and NGRDI) for analysis
- Built a Random Forest model for nutrient-deficiency classification

**Research Paper Ranking And Recommendation System (SIC Project)** May 2025
*Tools: Python, SciBERT, SBERT, OpenCV*

- Developed an AI-based research paper ranking and recommendation system
- Integrated OCR and computer vision for figure and table extraction
- Built a web interface for interactive literature analysis
- Applied NLP models (SciBERT and SBERT) for paper relevance evaluation

**Adaptive Drone Swarm Behavior Simulation** Jan 2025
*Tools: Python, NumPy, Matplotlib, Multi-Agent Simulation*

- Developed a simulation framework for adaptive drone swarm behavior
- Implemented coordinated movement and formation control among multiple drones
- Designed obstacle-avoidance and dynamic path-planning algorithms
- Evaluated swarm performance under varying environmental conditions

**Aerombulance - Ambulance Aircraft for Medical Operations** July 2025
*Tools: CATIA, XFLR5, Plane Maker, X-Plane*

- Designed a STOL (Short Takeoff And Landing) aircraft for remote and short-field operations
- Developed a modular cabin for patient and neonatal transport
- Defined mission profiles and system requirements for medical operations

## ADDITIONAL PROJECTS

**Autonomous UAV Navigation Using GPS and Pixhawk**

- Assembled a quadcopter using a Pixhawk flight controller
- Integrated GPS and configured autonomous waypoint missions
- Calibrated onboard sensors and flight-control parameters
- Performed test flights and basic flight performance evaluation

### UAV Mapping and Orthomosaic Generation

- Captured aerial imagery for UAV-based mapping applications
- Generated orthomosaic maps from drone image datasets
- Evaluated image overlap and survey coverage for mapping accuracy

### Real Time Face Detection on Tello Drone

*Tools: Python, OpenCV, DJI Tello SDK*

- Developed a real-time face detection system using OpenCV and Python
- Integrated the system with the DJI Tello video stream
- Tested live face detection during drone flights

### RC Fixed Wing Design and Flight Testing

- Designed and fabricated an RC fixed-wing aircraft for flight testing
- Integrated propulsion, control surfaces, and radio communication systems
- Carried out test flights to check flight performance
- Observed stability and control behavior during flight tests

### Drone Payload Dropping Mechanism

- Designed a lightweight payload release mechanism for a quadcopter
- Integrated a servo-based actuation system for payload deployment
- Tested the mechanism for controlled release during flight

## SKILLS

**Programming:** Python, MATLAB, C, Arduino

**Machine Learning & AI:** OpenCV, Scikit-learn, Random Forest, Computer Vision, Image Segmentation, Feature Extraction, NLP (SciBERT, SBERT)

**UAV & Aerospace:** Pixhawk Flight Controller, Autonomous Flight Setup, UAV Mapping, RC Plane Design

**Design & Mapping Software:** CATIA, QGIS, DroneDeploy

**Research Skills:** Data Collection, Data Analysis, Literature Review, Technical Report Writing (LaTeX)

## TRAININGS & CERTIFICATIONS

- Samsung Innovation Campus (SIC) Python Training
- CAN-SAT Training, Orion Space
- Work in Machine Learning for Agriculture Applications

## REFERENCES

**Asst. Prof. Kamal Darlami**
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**Asst. Prof. Swaviman Acharya**
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